

A 9-State Markov Chain Model for Semi-Fixed Menstruation Patterns: Stationary Distribution and Population Heterogeneity

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Abstract

Semi-fixed menstruation (*veset hatzi kavu'a*) is a central construct in Jewish Halachic law governing pre-menstrual abstinence. A woman enters a semi-fixed state when two consecutive menstrual cycles fall within a consistent range of lengths; once in this state she is obligated to observe pre-menstrual abstinence on the expected day of onset unless her range excludes the liturgically significant days (day 30 and the monthly Halachic date). Despite its legal importance, the long-run probabilistic behaviour of semi-fixed menstruation has not previously been studied formally.

We model the longitudinal dynamics of menstrual cycle lengths as a discrete-time, finite Markov chain with nine states, partitioned according to whether a cycle is the first, second, or third-or-later consecutive cycle within a semi-fixed stretch, and by whether the accumulated range is *short* (maximum < 29 days), *normal* ($29 \leq L \leq 31$ days), or *long* (minimum > 31 days). States S_{31} and S_{33} are the *favourable* semi-fixed states in which no abstinence at day 30 or at the monthly Halachic date is required.

Using a Natural Family Planning (NFP) dataset of 1,554 menstrual cycles from 118 clients (reduced to 1,221 cycles from 98 clients after excluding participants who had established a *Haflaga* fixed menstruation), we estimate the empirical transition matrix and derive the unique stationary distribution analytically via eigenvalue decomposition. The stationary probability of the two favourable states is $\pi_{S_{31}} + \pi_{S_{33}} \approx 8.94\%$, while the overall long-run probability of being in any semi-fixed state is approximately 63.19%.

We further stratify clients into three groups based on their highest-attained semi-fixed sub-type and show, using Kruskal-Wallis tests followed by Bonferroni-corrected Dunn post-hoc comparisons, that the three groups differ significantly in their cycle-length distributions ($H = 386.03$, $p < 10^{-84}$). Group-specific sub-chain analyses reveal that S_{31} -prone clients reach their favourable state in 18.6% of long-run cycles, while S_{33} -prone clients do so in 17.1%, both substantially above the population-average 8.94%. These results provide a rigorous probabilistic foundation for Halachic guidance and personalised cycle-based prediction tools.

Keywords: Markov chain; stationary distribution; menstrual cycle; Halachic law; semi-fixed menstruation; natural family planning; applied probability; population heterogeneity

1 Introduction

The menstrual cycle is a complex biological process governed by the hypothalamic–pituitary–ovarian axis, with cycle lengths that vary both within and between individuals [Treloar et al., 1967, Chiazze et al., 1968, Munster et al., 1992]. Accurate characterisation of this variability has applications in reproductive medicine, natural family planning (NFP), fertility treatment scheduling, and—in the context of the present work—Jewish religious law (Halacha).

In Halachic law, the laws of *Nidah* define periods of obligatory spousal separation following menstruation and govern the anticipation of the next onset. A central concept is the *fixed menstruation* (*veset kavu’a*): a pattern of onset sufficiently regular to impose anticipatory abstinence obligations. The literature distinguishes several types of fixed menstruation, including the *Haflaga* (three consecutive equal-length cycles), the *Dilug* (arithmetic progression of cycle lengths), and the weekly (*hashavua*) patterns [Ross, 2023, 2024b].

Among these, *semi-fixed menstruation* (*veset hatzi kavu’a*)—triggered after two consecutive cycles whose lengths fall within a shared range—has received the least quantitative attention despite being, as our data confirm, the predominant long-run state. A woman in a semi-fixed state may be obligated to observe abstinence on day 30 of the cycle (counting from the most recent onset) and on the monthly Halachic date, unless her accumulated cycle-length range entirely excludes those days. This makes the sub-type of semi-fixed state she occupies legally and practically significant.

Contributions and paper organisation

This paper makes the following contributions:

1. We introduce a nine-state Markov chain that fully captures the longitudinal semi-fixed menstruation dynamics, with a principled state-space that encodes both the ordinal position within a stretch and the range sub-type (Definition 2).
2. We estimate the empirical transition matrix from a real NFP dataset, derive the unique stationary distribution analytically (Proposition 1), and validate it by Monte Carlo simulation.
3. We show that the long-run probability of being in a *favourable* semi-fixed state (no extra Halachic obligations) is $\approx 8.94\%$, while that of being in *any* semi-fixed state is $\approx 63.19\%$.
4. We identify significant population heterogeneity across three client sub-groups and compute group-specific sub-chain stationary distributions that support personalised Halachic guidance.

The paper is organised as follows. Section 2 provides the necessary Halachic and mathematical background. Section 3 describes the dataset and pre-processing. Section 4 defines the nine-state Markov chain and the state assignment algorithm. Section 5 presents the transition matrix, stationary distribution, and hypothesis test results. Section 6 discusses the implications and limitations. Section 7 concludes.

2 Background

2.1 Halachic Framework

The Halachic interval (*Haflaga*) between two consecutive menstrual onsets equals the calendar cycle length plus one, because Jewish law counts the day of onset as the first day of the new cycle rather than the last day of the preceding one. Throughout this paper we use the Halachic convention: if the calendar cycle length (from the end of menstrual flow onset to the next onset) is L days, the Halachic *Haflaga* is $\tilde{L} = L + 1$.

Definition 1 (Semi-fixed menstruation). *A woman establishes a semi-fixed menstruation (veset hatzi kavu'a) when two consecutive Halachic cycle lengths $\tilde{L}_{n-1}$ and $\tilde{L}_n$ satisfy $[\min(\tilde{L}_{n-1}, \tilde{L}_n), \max(\tilde{L}_{n-1}, \tilde{L}_n)] \subseteq \mathbb{Z}_{>0}$. The range $[r_{\min}, r_{\max}]$ defines her semi-fixed interval. She remains semi-fixed as long as each subsequent cycle length falls within this interval, which can only widen (absorb a new extreme) or be exited entirely (if the new length falls outside).*

Once semi-fixed, she must observe anticipatory abstinence on day 30 of the current cycle and on the day of the month corresponding to her last onset, unless both of these days fall strictly outside her semi-fixed interval. This motivates the distinction between sub-types:

- **Short range** ($r_{\max} < 29$): day 30 and the monthly date both fall outside the interval — *no extra abstinence required.*
- **Normal range** ($r_{\min} \leq 30 \leq r_{\max}$ or the interval contains the monthly date): extra abstinence required on at least one day.
- **Long range** ($r_{\min} > 31$): day 30 falls outside the interval, relieving that obligation, but the monthly-date obligation may remain.

A woman exits the semi-fixed state when a new cycle length falls outside her current interval; she then restarts the two-cycle acquisition process.

2.2 Markov Chain Fundamentals

A discrete-time, finite, homogeneous Markov chain is a stochastic process $\{X_n\}_{n \geq 1}$ over a state space $\mathcal{S} = \{s_1, \dots, s_K\}$ satisfying the Markov property $\Pr(X_{n+1} = s_j \mid X_n = s_i, X_{n-1}, \dots) = p_{ij}$, where $\mathbf{P} = (p_{ij})$ is the $K \times K$ transition matrix with $p_{ij} \geq 0$ and $\sum_j p_{ij} = 1$ for all i [Norris, 1997].

If the chain is irreducible and aperiodic, it possesses a unique stationary distribution $\boldsymbol{\pi} = (\pi_1, \dots, \pi_K)$ satisfying $\boldsymbol{\pi} \mathbf{P} = \boldsymbol{\pi}$, $\pi_k > 0$, $\sum_k \pi_k = 1$ [Kemeny and Snell, 1960]. In practice, $\boldsymbol{\pi}$ is computed as the left eigenvector of $\mathbf{P}$ corresponding to the eigenvalue 1:

$$\mathbf{P}^\top \boldsymbol{\pi} = \boldsymbol{\pi}, \quad \|\boldsymbol{\pi}\|_1 = 1. \quad (1)$$

3 Data and Pre-processing

3.1 Dataset

We use an anonymised Natural Family Planning (NFP) dataset collected by certified NFP instructors. Each record corresponds to one menstrual cycle for one participant and includes the calendar cycle length, cycle number within the participant’s observation period, and demographic variables (age, BMI, parity, etc.).

The raw dataset contains 1,665 cycle records from 133 clients. After removing duplicate (ClientID, CycleNumber) pairs and a small number of records with known data anomalies (client 8107, cycles 45–53, identified during data collection), and restricting to participants with at least five recorded cycles to ensure sufficient longitudinal information, the cleaned dataset comprises **1,554 cycles from 118 clients**, with a mean of 13.2 cycles per client (median: 12, range: 5–45).

3.2 Halachic Cycle Length Convention

Following Definition 1, all analyses use the Halachic cycle length $\tilde{L} = L + 1$. Descriptive statistics of $\tilde{L}$ are shown in Table 1.

The distribution is right-skewed (skewness = 1.38) and leptokurtic (excess kurtosis = 5.71), consistent with prior studies of menstrual cycle length variability [Guo et al., 2006, Symul et al., 2019]. A Shapiro-Wilk test confirms non-normality ($W = 0.924$, $p < 10^{-30}$).

3.3 Exclusion of Haflaga-Fixed Participants

Participants who establish a *Haflaga* fixed menstruation (three consecutive equal Halachic cycle lengths) operate under a different Halachic regime and their cycles would introduce structural autocorrelations that violate the intended semi-fixed dynamics. We therefore

Table 1: Descriptive statistics of the Halachic cycle length $\tilde{L}$ across the filtered dataset ($n = 1,554$ cycles, 118 clients).

Statistic	Value
Number of cycles	1,554
Number of clients	118
Mean	30.27 days
Standard deviation	3.86 days
Median	29.50 days
Q1 – Q3	28 – 32 days
Minimum	19 days
Maximum	55 days
Skewness	1.38
Kurtosis (excess)	5.71

exclude the 20 participants who exhibited at least one *Haflaga* event, yielding the **analysis dataset of 1,221 cycles from 98 clients** used for all subsequent Markov chain modelling.

4 The 9-State Markov Chain Model

4.1 State Space

Definition 2 (State space $\mathcal{S}$). *We define nine states $\mathcal{S} = \{S_{11}, S_{12}, S_{13}, S_{21}, S_{22}, S_{23}, S_{31}, S_{32}, S_{33}\}$ for each cycle, where the first index encodes the ordinal position within the current stretch and the second index encodes the range sub-type:*

First index: $1 \Rightarrow$ first cycle of a new stretch (semi-fixed interval resets),
 $2 \Rightarrow$ second cycle (semi-fixed interval being formed),
 $3 \Rightarrow$ third or later cycle (semi-fixed state achieved).

Second index: $1 \Rightarrow r_{\max} < 29$ (short range),
 $2 \Rightarrow 29 \leq r_{\min}$ or $r_{\max} \leq 31$ (normal range),
 $3 \Rightarrow r_{\min} > 31$ (long range).

States $S_{3\bullet}$ are the *semi-fixed states*; among them, S_{31} and S_{33} are the *favourable states* requiring no additional abstinence at day 30 or at the monthly Halachic date.

4.2 State Assignment Algorithm

Algorithm 1 describes the sequential state assignment for a sequence of Halachic cycle lengths $(\tilde{L}_1, \tilde{L}_2, \dots)$ belonging to one client.

Algorithm 1 Semi-fixed state assignment for one participant

Require: Sequence $(\tilde{L}_1, \dots, \tilde{L}_N)$, threshold $\theta_{\text{short}} = 29$, $\theta_{\text{long}} = 31$

Ensure: State sequence $(X_1, \dots, X_N)$

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1:  $r_{\min} \leftarrow \tilde{L}_1$ ;  $r_{\max} \leftarrow \tilde{L}_1$ ;  $\text{sf} \leftarrow 0$  ▷  $\text{sf} = 1$ : currently semi-fixed
2:  $X_1 \leftarrow \text{SUBTYPE}(r_{\min}, r_{\max}, \text{prefix} = \text{S1})$ 
3: for  $i = 2, 3, \dots, N$  do
4:   if  $\text{sf} = 1$  and  $(\tilde{L}_i < r_{\min} \text{ or } \tilde{L}_i > r_{\max})$  then
5:     ▷ Exit semi-fixed: range broken, restart
6:      $r_{\min} \leftarrow \tilde{L}_i$ ;  $r_{\max} \leftarrow \tilde{L}_i$ ;  $\text{sf} \leftarrow 0$ 
7:      $X_i \leftarrow \text{SUBTYPE}(r_{\min}, r_{\max}, \text{prefix} = \text{S1})$ 
8:   else if  $i = 2$  or  $(\text{sf} = 0 \text{ and } X_{i-2} \text{ starts with S3})$  then
9:     ▷ Second cycle of a new stretch
10:     $r_{\min} \leftarrow \min(r_{\min}, \tilde{L}_i)$ ;  $r_{\max} \leftarrow \max(r_{\max}, \tilde{L}_i)$ ;  $\text{sf} \leftarrow 0$ 
11:     $X_i \leftarrow \text{SUBTYPE}(r_{\min}, r_{\max}, \text{prefix} = \text{S2})$ 
12:   else
13:     ▷ Third+ cycle in range, or extending semi-fixed
14:     $r_{\min} \leftarrow \min(r_{\min}, \tilde{L}_i)$ ;  $r_{\max} \leftarrow \max(r_{\max}, \tilde{L}_i)$ ;  $\text{sf} \leftarrow 1$ 
15:     $X_i \leftarrow \text{SUBTYPE}(r_{\min}, r_{\max}, \text{prefix} = \text{S3})$ 
16:   end if
17: end for
18: procedure  $\text{SUBTYPE}(r_{\min}, r_{\max}, \text{prefix})$ 
19:   if  $r_{\max} < \theta_{\text{short}}$  then return  $\text{prefix} + "1"$ 
20:   else if  $r_{\min} > \theta_{\text{long}}$  then return  $\text{prefix} + "3"$ 
21:   else return  $\text{prefix} + "2"$ 
22:   end if
23: end procedure
```

Remark 1. The range $[r_{\min}, r_{\max}]$ is monotonically widening within any stretch: once a cycle length extends the range, it cannot subsequently shrink. This ensures that the sub-type index can only transition $1 \rightarrow 2$ or $2 \rightarrow 3$ (and never backwards) within a single stretch, which reflects the Halachic principle that an established semi-fixed interval persists until explicitly broken.

4.3 State Diagram

Figure 1 shows the directed graph of the estimated Markov chain. The structure reveals two absorbing-like regions: once a participant enters a $S_{3\bullet}$ state, the chain either self-loops (with high probability) or returns to a $S_{1\bullet}$ state, never going directly to $S_{2\bullet}$. Similarly, $S_{1\bullet}$ states always transition to $S_{2\bullet}$ states (forming a two-step acquisition path into $S_{3\bullet}$), and $S_{2\bullet}$ states always transition to $S_{3\bullet}$ states. This block structure is a direct consequence of the Halachic definition.

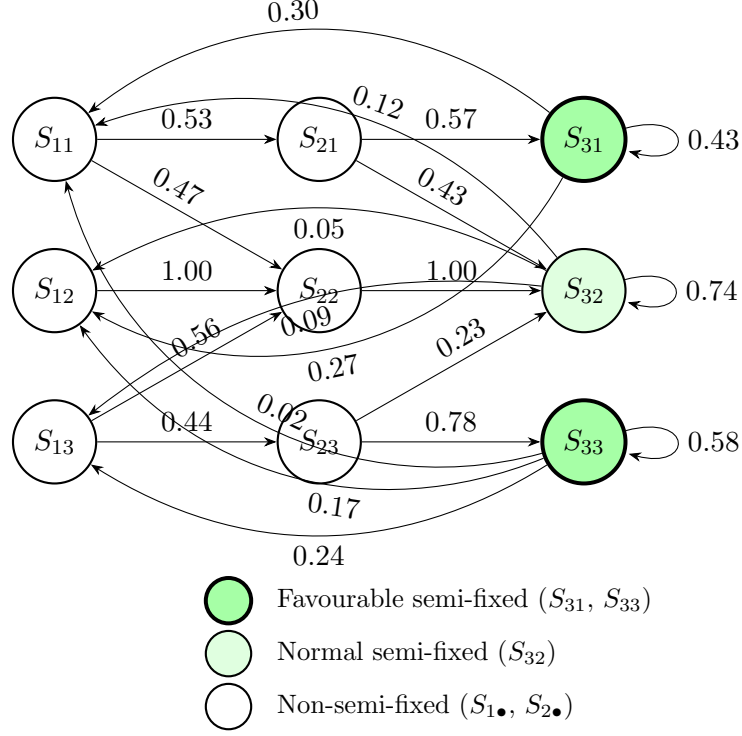


Figure 1: State transition diagram of the 9-state Markov chain (estimated transition probabilities shown on edges; values ≤ 0.01 omitted for clarity). Double-bordered states (S_{31}, S_{33}) are the *favourable* semi-fixed states in which no additional pre-menstrual abstinence is required.

4.4 Transition Matrix Estimation

We estimate the transition matrix $\mathbf{P}$ from observed consecutive-cycle pairs within each participant's sequence (transitions across participant boundaries are excluded). Let n_{ij} denote the observed count of transitions from state s_i to state s_j ; the maximum-likelihood estimate of each row is

$$\hat{p}_{ij} = \frac{n_{ij}}{\sum_k n_{ik}}.$$

Table 2 lists all non-zero transition counts; the corresponding probability matrix $\hat{\mathbf{P}}$ is reported in Table 3.

The block structure of $\hat{\mathbf{P}}$ is noteworthy:

- $S_{1\bullet}$ states transition exclusively to $S_{2\bullet}$ states (the sub-type of the destination may differ from that of the origin due to the range-widening rule).
- $S_{2\bullet}$ states transition exclusively to $S_{3\bullet}$ states.
- $S_{3\bullet}$ states either self-loop (with probability 0.43–0.74) or transition back to $S_{1\bullet}$ states (range broken).
- No state returns directly from $S_{3\bullet}$ to $S_{2\bullet}$, nor from $S_{2\bullet}$ to $S_{1\bullet}$.

Table 2: Observed transition counts n_{ij} (all entries not shown are zero).

From \ To	S_{11}	S_{12}	S_{13}	S_{21}	S_{22}	S_{23}	S_{31}	S_{32}	S_{33}	Total
S_{11}	—	—	—	48	42	—	—	—	—	90
S_{12}	—	—	—	—	76	—	—	—	—	76
S_{13}	—	—	—	—	54	42	—	—	—	96
S_{21}	—	—	—	—	—	—	27	20	—	47
S_{22}	—	—	—	—	—	—	—	161	—	161
S_{23}	—	—	—	—	—	—	—	9	31	40
S_{31}	13	12	—	—	—	—	19	—	—	44
S_{32}	59	26	45	—	—	—	—	373	—	503
S_{33}	1	11	16	—	—	—	—	—	38	66
Total	73	49	61	48	172	42	46	563	69	1,123

Table 3: Estimated transition probability matrix $\hat{\mathbf{P}}$ (entries < 0.001 shown as “—”).

From \ To	S_{11}	S_{12}	S_{13}	S_{21}	S_{22}	S_{23}	S_{31}	S_{32}	S_{33}
S_{11}	—	—	—	0.533	0.467	—	—	—	—
S_{12}	—	—	—	—	1.000	—	—	—	—
S_{13}	—	—	—	—	0.563	0.438	—	—	—
S_{21}	—	—	—	—	—	—	0.574	0.426	—
S_{22}	—	—	—	—	—	—	—	1.000	—
S_{23}	—	—	—	—	—	—	—	0.225	0.775
S_{31}	0.295	0.273	—	—	—	—	0.432	—	—
S_{32}	0.117	0.052	0.089	—	—	—	—	0.742	—
S_{33}	0.015	0.167	0.242	—	—	—	—	—	0.576

This three-level block structure is fully determined by the Halachic definition of semi-fixed menstruation and is a *model property*, not a data artefact.

5 Results

5.1 Stationary Distribution

Proposition 1 (Existence and uniqueness of $\boldsymbol{\pi}$). *The Markov chain with transition matrix $\hat{\mathbf{P}}$ is irreducible and aperiodic. Therefore, a unique stationary distribution $\boldsymbol{\pi}$ exists, satisfying $\boldsymbol{\pi}\hat{\mathbf{P}} = \boldsymbol{\pi}$ with $\pi_k > 0$ for all k .*

Sketch. Irreducibility follows by inspection of Table 3: every state communicates with every other state within at most four steps. Aperiodicity follows from the self-loops at S_{31} , S_{32} , and S_{33} (positive diagonal entries imply period 1). $\square$

We compute $\boldsymbol{\pi}$ as the normalised eigenvector of $\hat{\mathbf{P}}^\top$ corresponding to eigenvalue 1. The result is reported in Table 4 and illustrated in Figure 2.

Table 4: Stationary distribution π of the 9-state Markov chain.

State	π_k	Percentage
S_{11}	0.07656	7.66%
S_{12}	0.04732	4.73%
S_{13}	0.06019	6.02%
S_{21}	0.04083	4.08%
S_{22}	0.11690	11.69%
S_{23}	0.02633	2.63%
S_{31}	0.04128	4.13%
S_{32}	0.54248	54.25%
S_{33}	0.04811	4.81%
<i>Aggregates:</i>		
$\pi_{S_{31}} + \pi_{S_{33}}$ (favourable semi-fixed)	0.08939	8.94%
$\pi_{S_{31}} + \pi_{S_{32}} + \pi_{S_{33}}$ (all semi-fixed)	0.63186	63.19%

The dominant state is S_{32} with $\pi_{S_{32}} \approx 54.25\%$: in the long run, more than half of all cycles are spent in a normal-range semi-fixed state, imposing full abstinence obligations on day 30 and the monthly Halachic date. The two favourable states together account for only $\pi_{S_{31}} + \pi_{S_{33}} \approx 8.94\%$ of long-run cycles.

Remark 2. *The high stationary probability of S_{32} is a direct consequence of its large self-loop probability ($\hat{p}_{S_{32}, S_{32}} = 0.742$): once a participant enters the normal-range semi-fixed state, she is likely to remain there for many consecutive cycles. This persistence is driven by the absorbing tendency of typical (28–32 day) cycle lengths in the NFP population.*

Monte Carlo Validation

To validate the analytical result, we simulated the chain for $T = 50,000$ steps and compared the empirical visit frequencies with π . The maximum absolute deviation was 3.6×10^{-4} , confirming convergence to the analytical stationary distribution.

5.2 Population Heterogeneity: Group Analysis

We partition clients into three groups based on the highest-priority semi-fixed sub-type they entered during observation:

- **Group G_{31} :** clients who entered S_{31} at least once ($n = 16$ clients, $N_{31} = 243$ cycles);
- **Group G_{33} :** clients who entered S_{33} at least once and are not in G_{31} ($n = 24$ clients, $N_{33} = 304$ cycles);
- **Group G_{32} :** remaining clients, i.e., those whose semi-fixed cycles were always in S_{32} ($n = 58$ clients, $N_{32} = 674$ cycles).

9-State Markov Chain Analysis of Semi-Fixed Menstruation

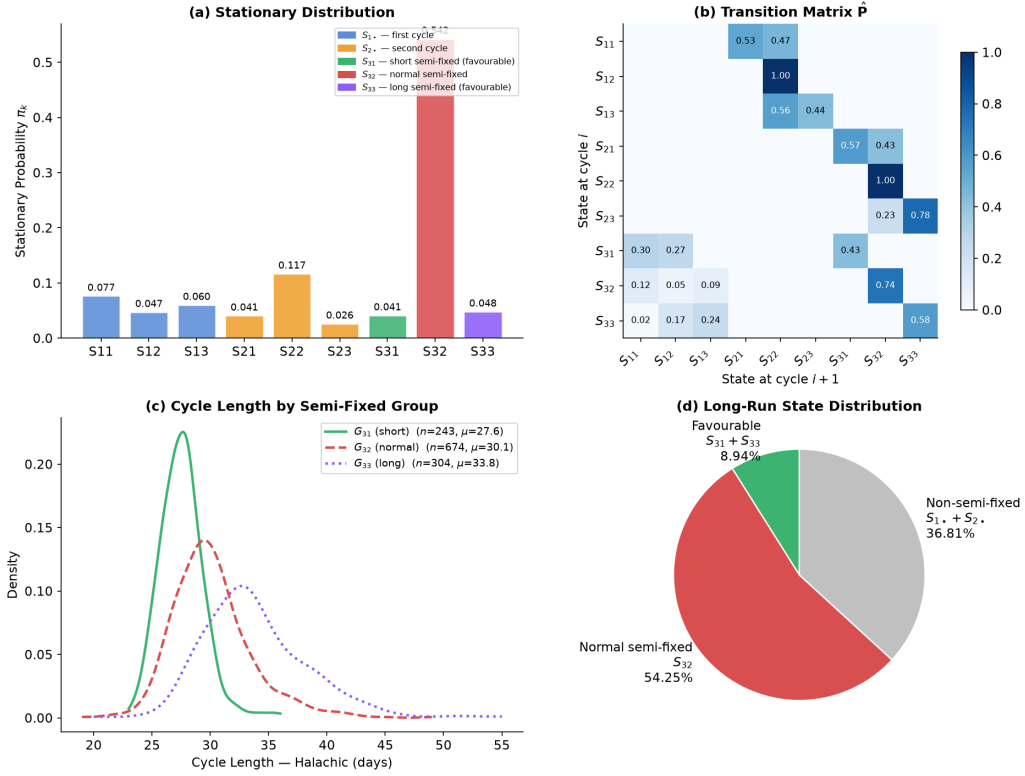


Figure 2: (a) Stationary distribution π over the nine states (green = favourable semi-fixed; red = normal semi-fixed; grey = non-semi-fixed). (b) Estimated transition matrix $\hat{P}$ as a heatmap. (c) Kernel density estimates of cycle lengths for the three semi-fixed client groups. (d) Long-run cycle state distribution (favourable vs. normal semi-fixed vs. non-semi-fixed).

Descriptive statistics per group are shown in Table 5. The clear stratification in mean cycle length— G_{31} clients have short cycles, G_{32} have normal cycles, and G_{33} have long cycles—is consistent with the state definitions.

Table 5: Descriptive statistics of Halachic cycle length $\tilde{L}$ by group (Halachic convention, $\tilde{L} = L + 1$).

Group	n (clients)	N (cycles)	Mean	SD	Median
G_{31} (short range)	16	243	27.55	1.87	27.0
G_{32} (normal range)	58	674	30.13	3.41	30.0
G_{33} (long range)	24	304	33.85	4.48	33.0

5.3 Hypothesis Testing

Normality

Shapiro-Wilk tests on the cycle-length distributions of each group all reject normality at $\alpha = 0.05$ (Table 6), motivating the use of non-parametric tests.

Table 6: Shapiro-Wilk normality test results for cycle lengths in each group.

Group	W	p -value	Normal?
G_{31}	0.936	8.80×10^{-9}	No
G_{32}	0.939	5.12×10^{-16}	No
G_{33}	0.944	2.22×10^{-9}	No

Homogeneity of Variances

Levene’s test rejects equality of variances across groups ($F = 40.48$, $p = 9.58 \times 10^{-18}$), confirming heteroscedasticity.

Kruskal-Wallis Test

The non-parametric Kruskal-Wallis test strongly rejects the null hypothesis of equal medians across the three groups:

$$H = 386.03, \quad p = 1.50 \times 10^{-84}.$$

Post-hoc Pairwise Comparisons

Dunn’s pairwise Kruskal-Wallis tests with Bonferroni correction (three comparisons, $\alpha_{\text{adj}} = 0.05/3 \approx 0.017$) confirm that all three pairs differ significantly (Table 7).

Table 7: Bonferroni-corrected Dunn pairwise test results. All comparisons are significant at $\alpha_{\text{adj}} = 0.017$.

Comparison	H	Adjusted p -value
G_{31} vs G_{32}	146.48	3.05×10^{-33}
G_{31} vs G_{33}	306.19	4.42×10^{-68}
G_{32} vs G_{33}	177.49	5.14×10^{-40}

5.4 Sub-Chain Analysis

We re-estimate the transition matrix and stationary distribution separately within each group’s cycle sequences. These *group-conditioned sub-chains* answer the question: for a woman whose cycle lengths characteristically place her in group G_k , what fraction of her long-run cycles will be spent in a favourable semi-fixed state?

Group G_{31}

The G_{31} sub-chain involves states $\{S_{11}, S_{12}, S_{13}, S_{21}, S_{22}, S_{31}, S_{32}\}$ (state S_{33} never arises in this group by construction). The stationary probability of the favourable state is

$$\pi_{S_{31}}^{(31)} \approx 18.6\%,$$

more than *four times* the population-level value of 4.1%.

Group G_{33}

Similarly, the G_{33} sub-chain yields

$$\pi_{S_{33}}^{(33)} \approx 17.1\%,$$

approximately *three and a half times* the population-level value of 4.8%.

Interpretation

The population-level stationary distribution π provides a global summary, but substantially underestimates the favourable-state probability for clients with short or long cycle patterns. Personalised Halachic guidance that leverages a woman’s individual cycle-length profile could therefore predict her semi-fixed sub-type with far greater accuracy than a population-average model.

6 Discussion

6.1 Probabilistic Interpretation

The main finding — that $\approx 63\%$ of cycles lie in some semi-fixed state in the long run, with $\approx 54\%$ specifically in S_{32} — implies that the semi-fixed regime is the *default* operating mode of the menstrual cycle in an NFP-charting population. Far from being a rare special case, semi-fixed menstruation is the norm. The Halachic obligations it entails (pre-emptive abstinence on day 30 and on the monthly date) therefore apply to the majority of cycles for most women.

The finding that only $\approx 8.9\%$ of long-run cycles are in truly favourable semi-fixed states (S_{31} or S_{33} , where no additional obligations apply) underscores the importance of individual cycle-length profiling: a short-cycle woman (mean $\tilde{L} \approx 27.6$) has nearly a 1-in-5 chance of being in the favourable S_{31} state, while the population average suggests only 1-in-11.

6.2 Comparison with Prior Work

Prior work by the present author introduced Markov chain models for *Haflaga* fixed menstruation [Ross, 2024a] and permutation tests for fixed-pattern frequencies [Ross, 2023]. The present model extends this framework to the more prevalent semi-fixed case, with a richer state space (nine vs. two states) and the additional dimension of sub-type heterogeneity.

Compared to the cycle-length forecasting literature [Guo et al., 2006, Symul et al., 2019, Bull et al., 2019], our model is not predictive in the time-series sense (we do not forecast the next cycle length); rather, it is characterisational: we describe the long-run occupancy distribution over Halachically meaningful states. The two approaches are complementary.

6.3 Limitations

1. **Dataset size.** The analysis dataset comprises 98 clients and 1,221 cycles after exclusions. While sufficient for stable transition matrix estimation on the major arcs, some transitions (especially those entering S_{11} from S_{33} and S_{23} self-loops) have low counts and should be interpreted cautiously.
2. **Population specificity.** The NFP dataset consists predominantly of women who chart their cycles for religious or family-planning purposes; this may introduce selection bias relative to the general population.
3. **Stationarity assumption.** We assume cycle-length distributions are stationary over time within each participant. Systematic within-woman trends (due to age, parity, or hormonal changes) are not modelled.
4. **Range sub-type boundaries.** The thresholds $\theta_{\text{short}} = 29$ and $\theta_{\text{long}} = 31$ are determined by Halachic rules; a data-driven approach might choose different boundaries and yield a different stationary distribution.

7 Conclusion

We have introduced a nine-state Markov chain model for the dynamics of semi-fixed menstruation in Halachic law and derived its unique stationary distribution from an empirical NFP dataset. The key results are:

1. In the long run, **63.2%** of menstrual cycles lie in a semi-fixed state, confirming that this is the dominant mode of the cycle.
2. Of these, **8.94%** are in favourable semi-fixed states (S_{31} or S_{33}) where no additional abstinence obligations arise — the remainder (S_{32} , $\approx 54.2\%$) carry full Halachic obligations.

3. Clients stratify naturally into three groups with significantly different cycle-length distributions (Kruskal-Wallis $H = 386.03$, $p < 10^{-84}$), and group-specific sub-chains show that short-cycle (G_{31}) and long-cycle (G_{33}) women spend $\approx 18.6\%$ and $\approx 17.1\%$ of their cycles in their respective favourable states — far above the population average.

These results provide a rigorous probabilistic foundation for Halachic guidance on semi-fixed menstruation and demonstrate that personalised, cycle-length-aware models outperform population-average predictions. Future work will incorporate Bayesian updating of the transition matrix from an individual’s observed cycle history, enabling real-time personalised predictions.

Author Contributions

D.R. conceived the study, designed the Markov chain model, performed all data analysis, and wrote the paper.

Funding

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Data Availability

The anonymised dataset and all analysis code (Jupyter notebooks) are available at <https://github.com/dvirross/PhD/tree/main/markov-chain-semi-fixed-menstruation>.

Conflicts of Interest

The author declares no conflict of interest.

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